

SLIDRONIX

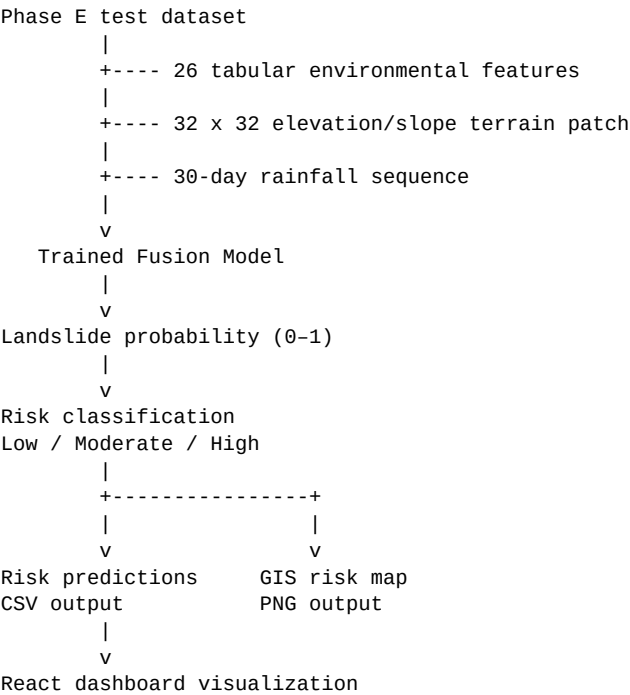
Phase H — Risk Mapping & Early Warning Implementation

Implementation checkpoint — September 2026

1. Phase H Objective

Phase H converts the trained multimodal Fusion model output into an interpretable spatial risk product. The implemented pipeline loads the trained Fusion checkpoint, aligns the Phase E test records with the valid CNN terrain patches and 30-day rainfall sequence, produces a landslide probability from 0 to 1, assigns a Low/Moderate/High risk level, and renders the predictions as a GIS-style map.

2. Implemented Phase H Pipeline



3. Model Used

The Phase H implementation uses the trained multimodal Fusion model developed in Phase F. The model contains three encoders: an MLP-based tabular encoder, a CNN spatial encoder, and a Transformer rainfall encoder. Their 32-dimensional embeddings are concatenated into a 96-dimensional fused representation and passed through dense layers to produce the final logit, which is converted to a probability.

Fusion components

Component	Input	Representation
Tabular encoder (MLP)	26 environmental features	32-dimensional embedding
CNN encoder	2 × 32 × 32 terrain patch	32-dimensional embedding
Rainfall Transformer	30 daily rainfall values	32-dimensional embedding

Fusion head	$32 + 32 + 32 = 96$	Landslide probability
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4. Phase H Step 1 — Risk Prediction

Script: **generate_phase_h_risk_predictions.py**. The implementation uses the trained Fusion checkpoint and applies the same preprocessing approach used during model training. Because CNN patches are unavailable for some locations, Phase H predictions are generated for the common aligned test subset.

Item	Implemented result
Phase E test rows	2,800
Valid CNN test samples	2,761
Aligned Phase H prediction samples	2,761
Probability range	0.0001 – 0.9978
Mean probability	0.4974

5. Risk Classification

Each predicted probability is converted into a categorical risk level used for dashboard and GIS visualization. The current implementation produces three classes: Low, Moderate and High.

Risk level	Count	Share of 2,761
Low	1,083	39.22%
Moderate	490	17.75%
High	1,188	43.03%

6. Phase H Step 2 — GIS Risk Map

Script: **create_phase_h_risk_map.py**. The generated map plots the 2,761 valid prediction points using latitude and longitude. Point symbols are colored according to the calculated risk level. The output is saved as a PNG suitable for reports and presentations.

Generated files

outputs/phase_h/slidronix_phase_h_risk_predictions.csv

outputs/phase_h/phase_h_risk_summary.txt

outputs/phase_h/slidronix_phase_h_risk_map.png

outputs/phase_h/phase_h_risk_map_summary.txt

7. Dashboard Integration

A React + Leaflet dashboard has been implemented to visualize the Phase H predictions. The dashboard uses satellite imagery as the default map layer, supports a street-map alternative, filters points by risk level, and allows the user to select prediction points. Selecting a point displays its risk level, probability, coordinates and selected environmental features.

Current dashboard flow

```
Risk prediction CSV
  |
  v
risk_predictions.json
  |
  v
React + Leaflet dashboard
  |
  +--> Satellite / street basemap
  +--> Risk-filtered prediction points
  +--> Selected-location analysis
  +--> Probability and feature display
```

8. Current Implementation Status

Phase H component	Status
Fusion model inference	Completed
Probability generation	Completed
Low/Moderate/High classification	Completed
GIS risk map	Completed
React dashboard	Completed
Point selection and location details	Completed
Dynamic coordinate → live environmental APIs → inference	Future enhancement
Real-time rainfall-triggered alerts	Future enhancement

9. Important Scope Note

The current Phase H implementation is a prepared-data risk visualization and model-inference stage. The rainfall sequence used by the current model is the fixed 1–30 January 2022 reference period established in Phase D; it is not an event-aligned real-time rainfall trigger. The next enhancement is to allow a user-selected coordinate to trigger retrieval of the latest available environmental inputs, feature construction and fresh Fusion-model inference.

10. One-line Viva Explanation

“Phase H takes the trained multimodal Fusion model, generates landslide probabilities for spatially aligned test locations, converts those probabilities into risk classes, and visualizes the resulting risk distribution through a GIS map and interactive dashboard.”

11. Main Implementation Files

File	Purpose
src/generate_phase_h_risk_predictions.py	Fusion-model inference and risk classification
src/create_phase_h_risk_map.py	GIS risk-map generation
outputs/phase_h/slidronix_phase_h_risk_predictions.csv	Spatial prediction output
outputs/phase_h/slidronix_phase_h_risk_map.png	GIS risk-map output
frontend/src/App.jsx	Interactive dashboard

frontend/src/App.css

Dashboard styling